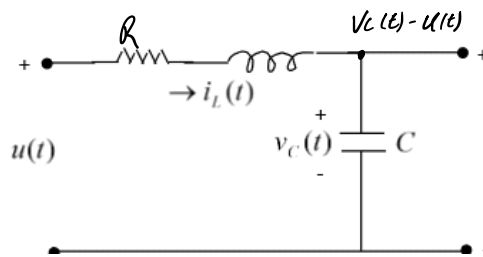


EEEE- 414: Classical Control

Homework Assignment # 3

Due date: by 11:59 PM, Thursday, September 17, 2026**Problem #1**

Given the circuit shown below, find the value of the resistor R such that a 20% overshoot in voltage will be seen across the capacitor if $C = 1\mu\text{f}$ and $L=1\text{H}$



$$Z = \frac{\ln(0\%/100)}{\sqrt{\tau^2 + (\ln(0\%/100))^2}}$$

$$Z = \frac{-\ln(20/100)}{\sqrt{\tau^2 + (\ln(20/100))^2}}$$

$$Z = 0.456$$

$$u(s) = I_c(R + sL + \frac{1}{sC})$$

$$v_C(s) = I_c \frac{1}{sC}$$

$$\frac{v_C(s)}{u(s)} = \frac{\frac{1}{sC}}{R + sL + \frac{1}{sC}}$$

$$\frac{v_C(s)}{u(s)} = \frac{1}{sRC + s^2LC + 1}$$

$$\frac{v_C(s)}{u(s)} = \frac{1/C}{s^2 + sR/L + 1/LC}$$

$$2\zeta\omega_n = R/L$$

$$R = 2\zeta\omega_n L$$

$$\omega_n = \frac{1}{\sqrt{LC}}$$

$$\omega_n = \frac{1}{\sqrt{1 \cdot 10^{-6}}}$$

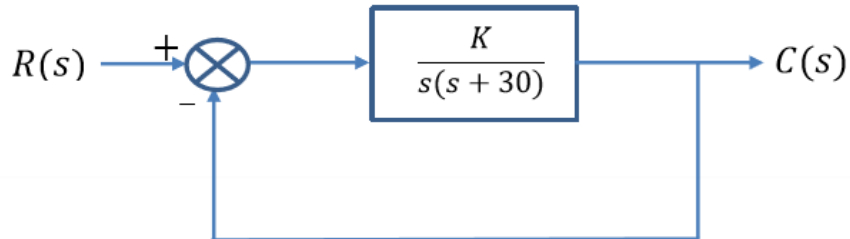
$$\omega_n = 1000$$

$$R = 2(0.456)(1000)(1)$$

$$R = 912 \text{ } \underline{R}$$

Problem # 2

For the system shown below, find the value of K that yields 10% overshoot for a step input.



$$G_{CL} = \frac{N(s)}{D(s) + N(s)}$$

$$G_{CL} = \frac{K}{s^2 + 30s + K}$$

$$Z = \frac{-\ln(0.9/100)}{\sqrt{\pi^2 + \ln(0.9/100)^2}}$$

$$Z = \frac{-\ln(0.9/100)}{\sqrt{\pi^2 + \ln(0.9/100)^2}}$$

$$Z = 0.591$$

$$2Z\omega_n = 30$$

$$\omega_n = 30/2Z$$

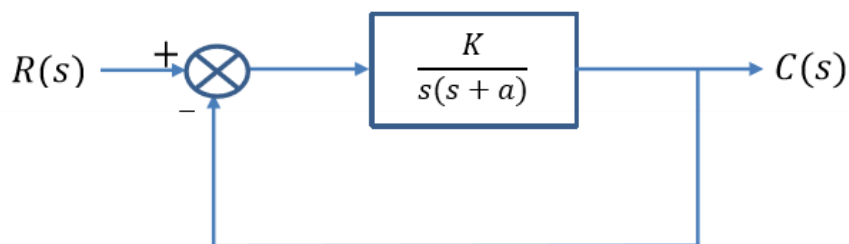
$$\omega_n = 25.374$$

$$\omega_n^2 = K$$

$$K = 643.843$$

Problem # 3

For the system shown below, find K and a to yield a settling time of 0.12[s] and a 20% overshoot for a step input.



$$G_{CL}(s) = \frac{K}{s^2 + as + K}$$

$$\frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$G_{cl}(s) = \frac{K}{s^2 + as + K}$$

$$\frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$\zeta = \frac{-\ln(0.05/100)}{\sqrt{\pi^2 + \ln(0.05/100)^2}}$$

$$\zeta = \frac{-\ln(2\%/100)}{\sqrt{\pi^2 + \ln(2\%/100)^2}}$$

$$\zeta = 0.456 \quad \text{- underdamped}$$

$$2\zeta\omega_n = a$$

$$T_s = 1/\zeta\omega_n$$

$$\omega_n = 1/T_s$$

$$\omega_n = 1/0.456(0.12)$$

$$\omega_n = 73.107$$

$$a = 2(0.456)(73.107)$$

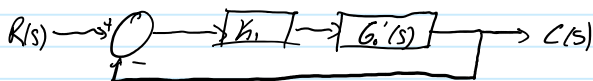
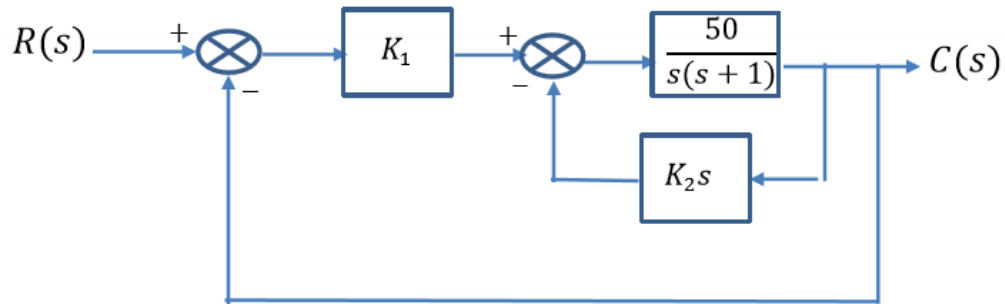
$$a = 66.67$$

$$\omega_n^2 = K$$

$$K = 5344.7$$

Problem # 4

For the system shown below, find the values of K_1 and K_2 to yield a peak time of 1 second and a settling time of 2 seconds for the closed loop system's step response.



$$G_o'(s) = \frac{G_o(s)}{1 + (H(s)G_o(s))}$$

$$G_o'(s) = \frac{\frac{50}{s(s+1)}}{1 + (K_2s) \frac{50}{s(s+1)}}$$

$$G_o'(s) = \frac{\frac{50}{s(s+1)}}{1 + \frac{50K_2}{s(s+1)}}$$

$$G_o'(s) = \frac{50}{s(s+1) + 50K_2s}$$

$$G_o'(s) = \frac{50}{s^2 + s + 50K_2s}$$

$$G_o'(s) = \frac{50}{s^2 + (1 + 50K_2)s}$$

$$G_{cl}(s) = \frac{N(s)}{D(s) + N(s)}$$

$$G_{cl}(s) = \frac{50K_1}{s^2 + (1 + 50K_2)s + 50K_1}$$

$$2\zeta\omega_n = 1 + 50K_2$$

$$\zeta = \frac{1 + 50K_2}{2\omega_n}$$

$$T_p = \frac{\pi}{\sqrt{1 - \zeta^2} \omega_n}$$

$$\omega_n = \sqrt{50K_1}$$

$$1 = \frac{\pi}{\sqrt{1 - \zeta^2}} \cdot \sqrt{50K_1}$$

$$\pi = \sqrt{1 - \zeta^2} \cdot \sqrt{50K_1}$$

$$\pi = \sqrt{1 - \left(\frac{1 + 50K_2}{2\sqrt{50K_1}}\right)^2} \cdot \sqrt{50K_1}$$

$$\pi = \sqrt{50K_1 - 50K_1 \left(\frac{1 + 50K_2}{2\sqrt{50K_1}}\right)^2}$$

$$\pi = \sqrt{50K_1 - 50K_1 \left(\frac{(1 + 50K_2)^2}{200K_1}\right)}$$

$$\pi = \sqrt{50K_1 - \frac{1}{4}(1 + 50K_2)^2}$$

$$T_s = \frac{4}{\zeta\omega_n}$$

$$\zeta = \frac{4}{2\omega_n}$$

$$\zeta\omega_n = 2$$

$$\frac{1 + 50K_2}{2\omega_n} \cdot \omega_n = 2$$

$$1 + 50K_2 = 4$$

$$50K_2 = 3$$

$$K_2 = 0.06$$

$$K_1 = 0.277$$